



PLANET 5

TOI-700 D

- Description:** Rocky planet orbiting in the habitable zone, where liquid water could exist on the surface. It is likely tidally locked.
- Host star:** TOI-700, a small, cool red dwarf about 40% the mass of the Sun, and unusually quiet with few damaging flares.
- System:** One of 4 known planets orbiting TOI-700, and the third one out from the star.
- Location:** Constellation Dorado, about 102 light-years from Earth. Far too faint for the naked eye (magnitude 13).
- Discovery:** In 2020 by NASA's TESS space telescope: the first habitable-zone planet TESS ever found.

Datasheet

EXOPLANET DATA		
	TEMPERATURE	-4 °C
	PRESSURE	Unknown
	ATMOSPHERE	Unknown
	GRAVITY	1.09 g
	PLANET RADIUS	
	ORBITAL PERIOD	
	SEMI-MAJOR AXIS	
	TRANSIT DEPTH	
	TRANSIT DURATION	

HOST STAR DATA		
	NAME	TOI-700
	SPECTRAL TYPE	M2.5V
	TEMPERATURE	3459 K
	DISTANCE	102 ly
	MASS	0.415 M _☉
	STAR RADIUS	0.421 R _☉
	AGE	1.5 Gyr
	APPARENT MAGNITUDE	13.15
	NUMBER OF PLANETS	4

SURVIVAL CHALLENGE

Tidal locking: permanent day and night sides with extreme temperature contrast

NOTES



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NOTES

PHYSICS

 PLANET RADIUS	
 AI-ASSISTANCE	
 REFLECTIONS	

MATHEMATICS

 ORBITAL PERIOD		 SEMI-MAJOR AXIS	
 AI-ASSISTANCE			
 REFLECTIONS			

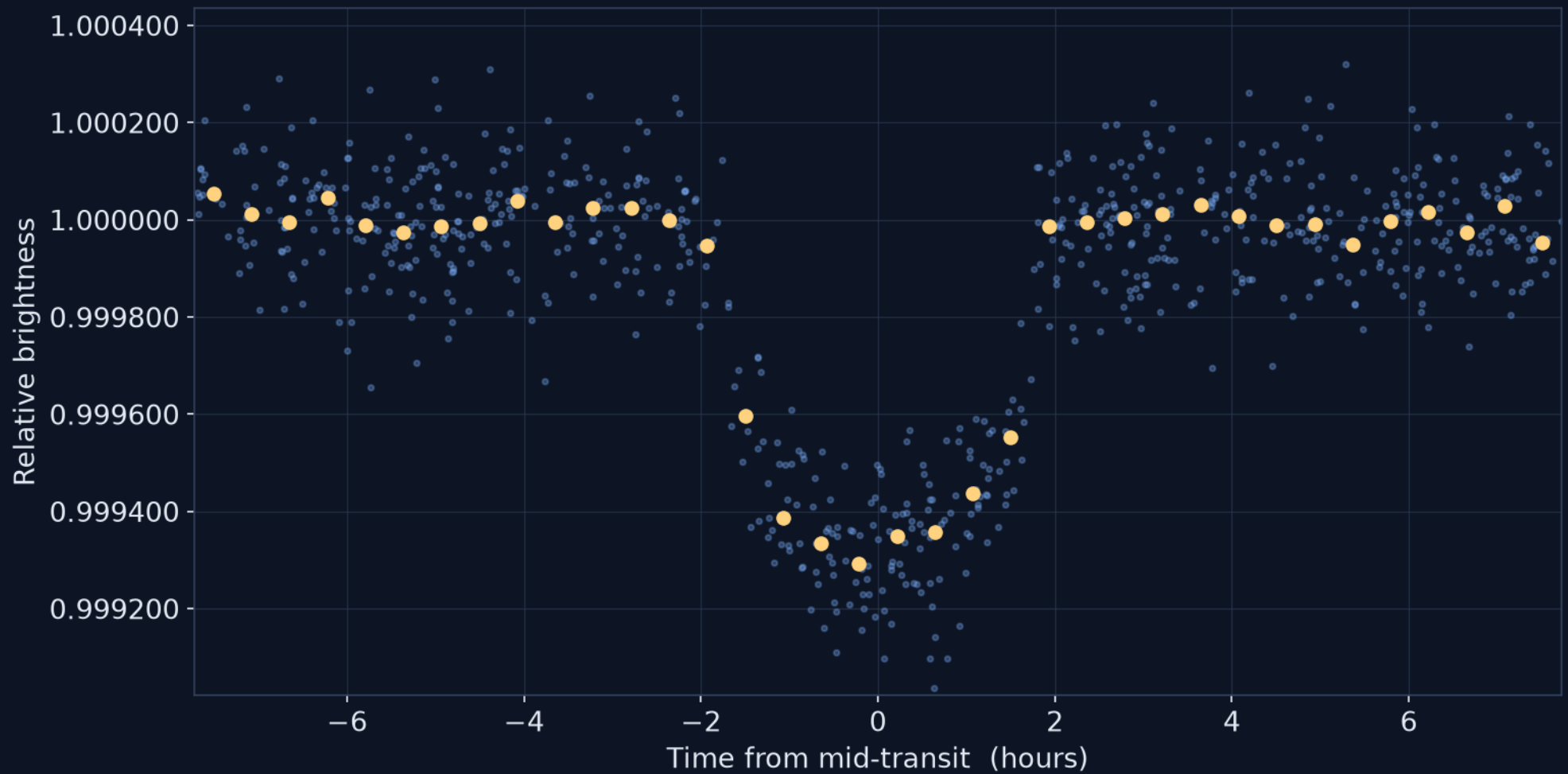
CHEMISTRY

 SAMPLE DATA	
 AI-ASSISTANCE	
 REFLECTIONS	

BIOLOGY

 ORGANISM	
 AI-ASSISTANCE	
 REFLECTIONS	

TOI-700 d: single transit



Blue points are simulated brightness measurements (many orbits phase-folded together, like real data); the gold points are binned averages.

TOI-700 d: light curve over several orbits

